SESE4143 · SOFTWARE RE-ENGINEERING · CLASS ACTIVITY

Why Did It Fail? Project Failure Analysis & Re-engineering SWOT

Each student researches ONE real software/tech project failure (2021–2026) from a global company and proposes a re-engineering solution.

The Task

- **Pick one** project from the list below. **No two students may pick the same project** — claim yours on the class sign-up sheet (first come, first served).
- **Research it yourself** using primary sources first (company post-mortems, regulator/government reports, court or inquiry documents), then reputable news and technical write-ups.
- Link the failure to **re-engineering concepts** from Weeks 1–3: legacy assessment, impact analysis, reverse engineering, refactoring, reuse, wrapping, migration, maintenance type.
- If the class is larger than the list, find your own qualifying project (failed *after* development/launch, last 5 years) and get it approved first — it must also be unique.

Your Deliverable — ONE Page (A4) + 2-Minute Talk

- **1. What failed** — what, when, who was affected (3–4 lines).
- **2. Why it failed** — technical and process causes, backed by cited sources.
- **3. Your re-engineering solution** — what you would change and how, using course terminology.
- **4. SWOT of <u>your</u> solution** — Strengths & Weaknesses (internal: cost, skills, effort, risk); Opportunities & Threats (external: market, regulation, technology, users).
- **5. Sources** — at least 3, with links.

S Strengths	W Weaknesses	O Opportunities	T Threats

Project Failures — Choose One (all launched or deployed after development; last 5 years)

1	**SpaceX Starlink**	Global network outage of ~2.5 hrs; internal software services failed (Jul 2025)
2	**Nvidia**	GeForce RTX 50-series launch drivers: black screens, crashes, repeated hotfixes (2025)
3	**Amazon AWS**	us-east-1 DynamoDB outage cascaded across dozens of services (Oct 2025)
4	**Cloudflare**	Network-wide failure to deliver core traffic worldwide (Nov 2025)
5	**CrowdStrike**	Falcon sensor update crashed ~8.5M Windows machines (Jul 2024)
6	**Microsoft**	Windows Recall AI feature pulled after security/privacy backlash (2024)
7	**Google**	Stadia cloud gaming shut down after ~3 years (announced 2022, ended 2023)
8	**Google**	Gemini image generation paused over inaccurate/biased outputs (2024)
9	**Meta**	Facebook, Instagram & WhatsApp down ~6 hours worldwide (Oct 2021)
10	**Meta**	Diem (Libra) digital currency project wound down (2022)
11	**Meta**	Horizon Worlds metaverse: adoption far below targets, heavy losses (2022–)
12	**Apple**	Project Titan self-driving car cancelled after ~10 years (2024)
13	**Apple**	Personalised Siri / Apple Intelligence features delayed after being advertised (2025)
14	**Amazon**	Halo health wearable and service discontinued (2023)
15	**Amazon**	"Just Walk Out" checkout scaled back in Fresh stores (2024)
16	**Zillow**	Zillow Offers pricing algorithm; unit wound down, huge losses (2021)
17	**IBM**	Watson Health / Watson for Oncology divested (2022)
18	**Southwest Airlines**	Crew-scheduling software meltdown; 16,000+ flights cancelled (Dec 2022)
19	**US FAA**	NOTAM system outage halted US departures nationwide (Jan 2023)
20	**Ticketmaster**	Eras Tour presale system collapsed under demand (Nov 2022)
21	**Sonos**	Rebuilt app launched missing core features; CEO stepped down (2024–25)
22	**Volkswagen**	Cariad in-house software unit: delayed EV launches (2022–24)
23	**Birmingham City Council**	Oracle ERP go-live: costs ballooned, finance/payroll problems (2022–)
24	**GM Cruise**	Robotaxi service suspended after incident, later shut down (2023–24)
25	**Humane**	AI Pin device discontinued ~10 months after launch (2025)
26	**Sony / Firewalk**	Concord game shut down two weeks after launch (2024)
27	**Intel**	13th/14th-gen Core CPUs: instability and crashes (2024)
28	**Optus**	Nationwide network outage affecting ~10M customers (Nov 2023)
29	**Rogers**	Canada-wide network and payments outage (Jul 2022)
30	**Twitter / X**	Paid "Blue" verification relaunch paused amid impersonation (Nov 2022)

Marking (10)

Failure analysis & sources	3	**SWOT quality**	2
Re-engineering solution	3	**One-page clarity + talk**	2

Ground Rules

- Original work; cite everything; copied or AI-generated text without your own analysis scores zero.
- Descriptions above are starting points only — verify every fact yourself.

Saira Latif · Lecturer (SE), FOIT CS · saira.latif@ucp.edu.pk · Office Hours 1:00–2:00 PM · University of Central Punjab